RIS-Assisted Learning-Based Anti-Jamming for Wideband THz Communications

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***Abstract*—Wideband terahertz (THz) communication systems are highly vulnerable to intelligent jamming due to frequency-selective propagation, beam squint, and highly directional transmissions. This paper proposes an intelligent reflecting surface-assisted anti-jamming framework for THz orthogonal frequency-division multiplexing systems that jointly optimizes transmit power allocation, reflecting-surface phase configuration, and hybrid beamforming. The resulting non-convex problem is addressed through a three-block reinforcement learning and alternating optimization framework. A fuzzy Win-or-Learn-Fast Policy Hill-Climbing algorithm is employed for adaptive power allocation against a smart jammer, while closed-form optimization updates are used for reflecting-surface configuration and beamformer design. To address wideband beam-squint effects, the proposed framework incorporates frequency-dependent reflecting-surface control across sub-arrays. Simulation results demonstrate that the proposed method achieves a system rate of 11.87 bits/s/Hz and a signal-to-interference-plus-noise ratio (SINR) protection level of 81.6%, yielding improvements of up to 44% in throughput and 66% in protection compared with representative reinforcement learning and deterministic optimization baselines. These results highlight the potential of combining intelligent reflecting surfaces and adaptive learning for robust wideband THz anti-jamming communications.**

***Index Terms*—Terahertz communications, intelligent reflecting surfaces, anti-jamming communications, reinforcement learning, hybrid beamforming, beam squint, physical layer security.**

I. Introduction

Terahertz (THz) communication has recently emerged as a promising technology for sixth-generation (6G) wireless networks due to the availability of ultra-wide bandwidth capable of supporting extremely high data rates. By exploiting spectrum resources beyond conventional millimeter-wave frequencies, THz systems can enable bandwidth-intensive applications such as immersive communications, holographic services, and integrated sensing and communication. However, THz propagation suffers from severe path loss, molecular absorption, and sparse scattering, necessitating highly directional beamforming and large antenna arrays to maintain reliable communication links [1], [2].

The reliance on directional transmission does not eliminate security vulnerabilities. In particular, intentional jamming remains a major threat to wireless networks, as adversaries can disrupt communication quality and significantly degrade system performance. Traditional anti-jamming techniques, such as frequency hopping and adaptive transmission strategies, have demonstrated effectiveness in conventional wireless systems [3], [4]. However, the unique propagation characteristics of THz channels, including severe path loss, molecular absorption, and frequency-selective fading, introduce additional challenges that complicate reliable communication and interference mitigation [5].

Recently, ~~intelligent reflecting surfaces (IRS) has attracted~~ considerable attention as a cost-effective solution for enhancing wireless coverage, spectral efficiency, and physical-layer security [6]–[8]. By intelligently adjusting the phase responses of passive reflecting elements, IRS can reshape the wireless environment and create favorable propagation conditions. IRS-assisted anti-jamming techniques have been widely studied to enhance communication resilience against interference. Reinforcement learning enables effective optimization of IRS-assisted anti-jamming strategies in dynamic environments with unknown and time-varying jammer behavior [9].

Despite these advances, several important research challenges remain. Most existing IRS-assisted anti-jamming studies focus on narrowband communication systems and do not explicitly account for the frequency-selective nature of wideband THz channels [9]. Beam-squint effects arising from large bandwidths and massive antenna arrays are often overlooked, although they can significantly degrade beamforming performance and reduce achievable gain [1], [2]. In addition, many reinforcement learning-based anti-jamming schemes rely on deterministic or highly predictable policies, which can be exploited by intelligent adversaries capable of inferring transmission behavior from historical observations. Furthermore, the joint optimization of power allocation, IRS configuration, and hybrid beamforming under adaptive jamming has not been comprehensively addressed in wideband THz systems.

These limitations motivate an integrated anti-jamming framework that combines IRS-assisted propagation control, beam-squint-aware wideband beamforming, and adaptive learning-based decision making for robust operation against intelligent and adaptive jammers. This paper proposes an IRS-assisted anti-jamming framework for wideband THz orthogonal frequency-division multiplexing (OFDM) systems that jointly optimizes transmit power allocation, IRS phase configuration, and sub-connected hybrid beamforming under adversarial conditions. A hybrid reinforcement learning and alternating optimization strategy is developed, where reinforcement learning handles discrete power control while closed-form updates are employed for IRS phase and beamformer optimization. To further enhance robustness, a fuzzy Win-or-Learn-Fast Policy Hill-Climbing (F-WoLF-PHC) algorithm is introduced to generate adaptive mixed transmission strategies that reduce policy predictability, while a beam-squint-aware IRS architecture is incorporated to maintain consistent wideband beamforming performance across subcarriers.

The main contributions of this paper are summarized as follows:

- We formulate a joint optimization problem for IRS-assisted anti-jamming in wideband THz communication systems, considering transmit power allocation, IRS phase design, and sub-connected hybrid beamforming under adaptive jamming.

- We develop a three-block reinforcement learning and alternating optimization framework that efficiently solves the resulting non-convex problem by combining learning-based power allocation with closed-form optimization of IRS and beamforming variables.
- We propose a F-WoLF-PHC algorithm that integrates fuzzy state aggregation and mixed-strategy learning to improve robustness against policy-exploiting intelligent jammers.
- We integrate an SPDP-RIS beam-squint compensation architecture (adapted from [1]) into the joint RL–AO anti-jamming framework, extending it with a multi-user AoD candidate search that re-optimizes the wideband RIS configuration jointly with the RL agent's instantaneous power allocation.
- Extensive simulations demonstrate significant improvements in system throughput and signal-to-interference-plus-noise ratio (SINR) protection compared with conventional reinforcement learning and deterministic optimization baselines.

II. System Model and Problem Formulation

A. Network Architecture

We consider a downlink wideband THz OFDM system comprising a base station (BS), an IRS, multiple single-antenna users, and an adaptive jammer. The BS is equipped with N antennas and N_{RF} RF chains, employing a sub-connected hybrid analog-digital architecture. The IRS consists of M reflecting elements arranged in a planar array to assist signal propagation between the BS and users. The system operates at a carrier frequency f_c with bandwidth B, divided into M_{sc} OFDM subcarriers. The BS serves K users while a multi-antenna jammer attempts to degrade system performance by injecting interference into the network. The IRS is deployed to enhance the desired signal strength and suppress jamming interference by intelligently adjusting its phase response.

Let $\mathbf{G}[m] \in \mathbb{C}^{M\times N}$ denote the BS-to-IRS channel on subcarrier m, $\mathbf{h}_{ru,k}[m] \in \mathbb{C}^{M}$ the IRS-to-user-k channel, and $\mathbf{h}_{bu,k}[m] \in \mathbb{C}^{N}$ the direct BS-to-user-k channel. The jammer-to-user channel is denoted by $\mathbf{h}_{J,k}$. Further, the IRS reflection matrix on subcarrier m is defined as $\mathbf{\Phi}[m] = \mathrm{diag}\left(e^{j\theta_1[m]}, \dots, e^{j\theta_M[m]}\right)$, where $\theta_i[m]$ represents the phase shift of the i-th reflecting element. The effective channel between the BS and user k is expressed as $\mathbf{h}_{\mathrm{eff},k}[m] = \mathbf{G}^H[m]\mathbf{\Phi}^H[m]\mathbf{h}_{ru,k}[m] + \mathbf{h}_{bu,k}[m]$. We assume the BS has perfect instantaneous channel state information (CSI) of $\mathbf{G}[m]$, $\mathbf{h}_{ru,k}[m]$, and $\mathbf{h}_{bu,k}[m]$ for IRS and beamformer design, a common idealization used to isolate the achievable performance of the proposed RL–AO control from channel-estimation error. The jammer's channel and strategy remain unknown to the BS and are inferred only indirectly, through SINR feedback and observed action predictability (Sec. III-F); robustness to imperfect/quantized BS-side CSI is left to future work.

B. THz Channel Model

THz channels exhibit strong frequency selectivity due to molecular absorption and beam-squint effects. The BS-RIS channel on subcarrier m is modeled using a geometry-based Saleh–Valenzuela representation [1]:

$$\mathbf{G}[m] = \sqrt{\frac{NM}{L_{\mathrm{path}}}} \sum_{\ell=1}^{L_{\mathrm{path}}} \alpha_\ell(f_m)\, \mathbf{a}_{\mathrm{RIS}}(\phi_\ell, f_m)\mathbf{a}_{\mathrm{BS}}^H(\psi_\ell, f_m) \tag{1}$$

where L_{path} is the number of propagation paths, $\alpha_\ell(f_m)$ is the complex path gain at subcarrier frequency $f_m = f_c + (m - M_{\mathrm{sc}}/2)\Delta f$ with subcarrier spacing $\Delta f = B/M_{\mathrm{sc}}$, and $\mathbf{a}_{\mathrm{RIS}}$, $\mathbf{a}_{\mathrm{BS}}$ are frequency-dependent array steering vectors. The frequency dependence of the steering vectors is the root cause of beam squint, as the spatial angle-of-arrival/departure at which constructive interference occurs shifts with frequency.

The THz path loss at frequency f and distance d is

$$\mathrm{PL}(f,d) = \left(\frac{4\pi f d}{c}\right)^2 e^{\kappa_{\mathrm{abs}}(f)d} \tag{2}$$

where c is the speed of light and $\kappa_{\mathrm{abs}}(f)$ is the molecular absorption coefficient. Unlike microwave channels, $\kappa_{\mathrm{abs}}(f)$ varies significantly within the 10 GHz bandwidth, introducing per-subcarrier attenuation diversity that the OFDM system must accommodate.

C. SPDP-RIS Architecture for Beam-Squint Compensation

To maintain beam alignment across the wideband THz spectrum, the RIS employs a sub-array phase-dependent partial-connection (SPDP) architecture [1]. The RIS is partitioned into $Q = 64$ sub-arrays, each equipped with a true-time-delay (TTD) unit and phase shifters. The reflection phase of element i in sub-array q is

$$\theta_{q,i}[m] = \theta_{q,i}^{\mathrm{base}} + 2\pi f_m \tau_q \tag{3}$$

where $\theta_{q,i}^{\mathrm{base}}$ is the base phase and τ_q is the TTD delay for sub-array q. This keeps beam direction consistent across subcarriers and avoids edge-subcarrier gain loss.

D. Signal Model

The received signal at user k on subcarrier m is given by

$$\begin{aligned} y_k[m] = {} & \mathbf{h}_{\mathrm{eff},k}^H[m]\mathbf{w}_k[m]\sqrt{P_k}s_k + \sum_{i\neq k}\mathbf{h}_{\mathrm{eff},k}^H[m]\mathbf{w}_i[m]\sqrt{P_i}s_i \\ & + \mathbf{h}_{J,k}^H\mathbf{z}_k\sqrt{P_{J,k}}j_k + n_k, \end{aligned} \tag{4}$$

where $\mathbf{w}_k[m]$ is the hybrid beamforming vector, P_k is the transmit power allocated to user k, s_k is the transmitted symbol, $P_{J,k}$ and $\mathbf{z}_k$ denote the jammer power and spatial precoder, respectively, and n_k is additive white Gaussian noise with variance σ^2. The SINR of user k on subcarrier m is defined as

$$\mathrm{SINR}_k[m] = \frac{P_k|\mathbf{h}_{\mathrm{eff},k}^H[m]\mathbf{w}_k[m]|^2}{\sum_{i\neq k}P_i|\mathbf{h}_{\mathrm{eff},k}^H[m]\mathbf{w}_i[m]|^2 + P_{J,k}|\mathbf{h}_{J,k}^H\mathbf{z}_k|^2 + \sigma^2}. \tag{5}$$

The achievable rate of user k is

$$R_k = \frac{1}{M_{\mathrm{sc}}}\sum_{m=1}^{M_{\mathrm{sc}}} \log_2\left(1 + \mathrm{SINR}_k[m]\right), \tag{6}$$

and the sum rate is $R_{\text{sum}} = \sum_{k=1}^{K} R_k$. To quantify quality-of-service (QoS) satisfaction under jamming, the SINR protection ratio is defined as

$$\eta_{\text{prot}} = \frac{1}{KM_{\text{sc}}} \sum_{k=1}^{K} \sum_{m=1}^{M_{\text{sc}}} \mathbf{1}\{\text{SINR}_k[m] \geq \gamma_{\min}\}, \tag{7}$$

where $\gamma_{\min}$ is the minimum SINR threshold.

E. Hybrid Beamforming

To balance hardware complexity and beamforming capability in the THz band, the BS employs a sub-connected hybrid analog–digital architecture with N antennas and N_{RF} RF chains. In this architecture, each RF chain is connected to a dedicated subset of $(N_{\text{per}} = N/N_{\text{RF}})$ antennas through analog phase shifters, resulting in a block-diagonal analog precoder. The transmit beamforming vector for user k on subcarrier m is expressed as

$$\mathbf{w}_k[m] = \mathbf{F}_{\text{RF}}\mathbf{f}_{BB,k}[m] \tag{8}$$

where $\mathbf{F}_{\text{RF}} \in \mathbb{C}^{N \times N_{\text{RF}}}$ is the frequency-flat analog precoder (block-diagonal, constant-modulus entries) and $\mathbf{f}_{BB,k}[m] \in \mathbb{C}^{N_{\text{RF}}}$ is the per-subcarrier digital precoder. The digital precoder adopts a max-SINR (Minimum Variance Distortionless Response (MVDR)) design. For user k at subcarrier m, the interference-plus-noise covariance is

$$\mathbf{R}_k[m] = \sigma^2\mathbf{I} + \sum_{i \neq k} P_i\, \tilde{\mathbf{h}}_i[m]\tilde{\mathbf{h}}_i^H[m] \tag{9}$$

where $\tilde{\mathbf{h}}_k[m] = \mathbf{F}_{\text{RF}}^H \mathbf{h}_{\text{eff},k}[m]$ is the compressed effective channel. The MVDR precoder is

$$\mathbf{f}_{BB,k}[m] = \frac{\mathbf{R}_k^{-1}[m]\tilde{\mathbf{h}}_k[m]}{\|\mathbf{R}_k^{-1}[m]\tilde{\mathbf{h}}_k[m]\|}. \tag{10}$$

F. Joint Optimization Problem

The objective is to maximize the system sum rate by jointly optimizing transmit power allocation, IRS phase shifts, and hybrid beamforming under QoS and hardware constraints. The optimization problem is formulated as

$$\max_{\{P_k\},\{\mathbf{\Phi}[m]\},\mathbf{F}_{\text{RF}},\{\mathbf{f}_{\text{BB},k}[m]\}} \quad R_{\text{sum}} \tag{11a}$$

$$\text{s.t.} \quad \sum_{k=1}^{K} P_k \leq P_{\max}, \tag{11b}$$

$$\text{SINR}_k[m] \geq \gamma_{\min},\ \forall k, m, \tag{11c}$$

$$|[\mathbf{\Phi}[m]]_{i,i}| = 1,\ \forall i, m, \tag{11d}$$

$$\mathbf{F}_{\text{RF}} \in \mathcal{F}_{\text{RF}}, \tag{11e}$$

$$\|\mathbf{F}_{\text{RF}}\mathbf{f}_{\text{BB},k}[m]\|^2 \leq 1,\ \forall k, m, \tag{11f}$$

where $\mathcal{F}_{\text{RF}}$ denotes the set of sub-connected constant-modulus analog precoders imposed by the RF hardware architecture.

The formulated problem is highly non-convex due to the coupling among power allocation, IRS phase shifts, and hybrid beamforming variables, as well as the unit-modulus and constant-modulus constraints. Moreover, the presence of an adaptive jammer with unknown strategy further complicates the optimization, making conventional deterministic approaches ineffective. Therefore, a hybrid reinforcement learning and alternating optimization framework is developed to efficiently solve the problem.

TABLE I
POWER-ALLOCATION ACTION SPACE ($7 \times 6 = 42$ ACTIONS)

Total power fraction $P_{\text{tot}}/P_{\max} \in \{0.20, 0.35, 0.50, 0.65, 0.80, 0.90, 1.00\}$	
Allocation mode	**Per-user weight**
Equal	uniform, $1/K$ each
Channel-proportional	$\propto$ channel quality
Inverse-channel	$\propto 1/$(channel quality)
SINR-deficit	$\propto \max(0, \gamma_{\min} - \text{SINR}_k)$
Water-filling	$\propto \log(1+$channel quality)
Max-protection	$\propto 1/\text{SINR}_k$ (worst-user focus)

III. PROPOSED RL–AO ANTI-JAMMING FRAMEWORK

The optimization problem in (11a)–(11f) is non-convex due to the coupling among transmit power allocation, RIS phase control, and hybrid beamforming variables. Furthermore, the jamming strategy is adaptive and unknown a priori, making model-based optimization impractical. To address these challenges, we develop a hybrid reinforcement learning and alternating optimization (RL–AO) framework in which reinforcement learning determines the transmit power allocation while closed-form alternating optimization updates the RIS phases and hybrid beamformer.

A. RL–AO Problem Decomposition

The optimization variables in (11a)–(11f) are partitioned into three blocks: transmit power allocation P_k, IRS phase shifts $\mathbf{\Phi}[m]$, and hybrid beamforming variables $\mathbf{F}RF, \mathbf{f}BB, k[m]$. Based on this structure, we develop a hybrid RL–AO framework in which reinforcement learning determines the power-allocation policy, while alternating optimization updates the IRS and beamforming variables. This decomposition reduces computational complexity and improves robustness against adaptive jamming. The resulting optimization is performed over the following three variable blocks:

1) **Block 1 — Power allocation (RL):** The RL agent selects a power allocation action from $\mathcal{A}$ ($|\mathcal{A}| = 42$ actions), formed as the Cartesian product of 7 total-power fractions and 6 per-user allocation modes (Table I). This block handles the combinatorial structure and model-free nature of the adversarial environment.
2) **Block 2 — IRS phase shifts (closed-form AO):** Given $\{P_k\}$ and $\mathbf{F}_{\text{RF}}$, the IRS phases that maximize aggregate desired signal power are computed in closed form per element, eliminating the need for iterative solvers.
3) **Block 3 — Hybrid beamforming:** Given $\{P_k\}$ and $\{\mathbf{\Phi}[m]\}$, the analog precoder is updated by SVD-based dominant-subspace projection under the constant-modulus constraint, and the digital precoder is re-solved via MVDR (10).

For a selected mode, per-user weights are computed from the current channel quality or SINR state and normalized to sum to one (via simplex projection) before being scaled by the chosen power fraction; this ensures every action satisfies constraint (11b) by construction.

The three blocks are updated in a hierarchical manner. At each environment step, the RL agent first selects a power-allocation action, after which the IRS phases and hybrid beamforming variables are updated via AO. This separation enables the learning module to focus on adversarial decision-making, while the continuous variables are optimized using instantaneous channel state information.

The IRS and beamforming updates are performed for $N_{\mathrm{AO}} = 3$ inner iterations per RL step to ensure fast tracking of the RL policy with low computational overhead. The power-allocation policy is updated at the end of each episode through interaction with the environment.

1) Block 2–IRS Phase Optimization via AO: Given $\{P_k\}$, the IRS phase configuration is updated in closed form via the SPDP design of Sec. II-C (3): for a chosen reference angle-of-departure, per-element phases align the AoA and AoD wavefronts across each sub-array, with the residual frequency-dependent phase compensated by the sub-array TTD delay τ_q. The reference AoD (and hence the SPDP configuration) is re-optimized every N_{AO} steps by evaluating $K+1$ candidate directions (the per-user AoDs and their centroid) against the achieved system rate under the current $\{P_k\}$ and $\mathbf{F}_{\mathrm{RF}}$, and selecting the direction that maximizes R_{sum}, ensuring wideband beam-squint compensation.

2) Block 3–Analog Precoder Update via SVD: Since $\mathbf{F}_{\mathrm{RF}}$ is frequency-flat, we aggregate the effective channel over reference subcarriers $\mathcal{M}_{\mathrm{ref}}$ as $\overline{\mathbf{H}} = \sum_{m\in\mathcal{M}_{\mathrm{ref}}} \mathbf{H}_{\mathrm{eff}}^{H}[m]$. For sub-array p, the row partition $\overline{\mathbf{H}}_p$ is SVD-decomposed and the analog column is updated as $[\mathbf{F}_{\mathrm{RF}}]_p = \exp(j\angle\mathbf{u}_p)/\sqrt{N_{\mathrm{per}}}$ (unit-modulus projection), then $\{\mathbf{f}_{BB,k}[m]\}$ is re-solved via MVDR (10).

B. State Representation

The system state is defined as a compact feature vector $\mathbf{f} = [f_{\mathrm{pj}}, f_{\mathrm{ch}}, f_{\mathrm{sinr}}] \in [0,1]^3$, capturing jammer activity, channel condition, and SINR quality, respectively.

The jammer pressure feature is defined as

$$f_{\mathrm{pj}} = \mathrm{norm}_{[15,40]}\left(0.6\,\overline{P_J} + 0.4\,\max_k P_{J,k}\right)\ [\mathrm{dBm}], \qquad (12)$$

where $\overline{P_J}$ denotes the average jammer power.

The channel quality indicator is given by

$$f_{\mathrm{ch}} = \mathrm{norm}_{[-100,-40]}\left(0.75\,\overline{\|\mathbf{h}_k\|^2} + 0.25(1-\mathrm{spread})\right)\ [\mathrm{dB}], \qquad (13)$$

while the SINR health metric is defined as

$$f_{\mathrm{sinr}} = \mathrm{norm}_{[-10,30]}\left(0.5\,\overline{\mathrm{SINR}_k} + 0.5\,\min_k \mathrm{SINR}_k\right)\ [\mathrm{dB}]. \qquad (14)$$

For tabular baselines, each feature is uniformly quantized into $B_q = 8$ levels, resulting in a discrete state space of size $8^3 = 512$. In contrast, the proposed WoLF-PHC framework operates on continuous inputs using fuzzy state aggregation to avoid discretization loss and improve learning stability.

C. Fuzzy State Aggregation

Each feature $x \in [0,1]$ is mapped to three triangular memberships with centers $\{0, 0.5, 1.0\}$:

$$\mu_i(x) = \frac{\max(0, 1 - |x - c_i|/\Delta)}{\sum_j \max(0, 1 - |x - c_j|/\Delta)} \qquad (15)$$

where $\Delta = 0.5$ is the spread parameter. The joint fuzzy membership vector has $3^3 = 27$ components:

$$\psi_\ell(\mathbf{f}) = \mu_i(f_{\mathrm{pj}}) \cdot \mu_j(f_{\mathrm{ch}}) \cdot \mu_k(f_{\mathrm{sinr}}), \quad \ell = 9i + 3j + k. \qquad (16)$$

This smooths state transitions and improves robustness.

D. Proposed Fuzzy WoLF-PHC Algorithm

WoLF-PHC [14] learns faster when losing and slower when winning. The fuzzy version maintains $Q_\ell(s,a)$, policy $\pi_\ell(a)$, and average policy $\bar{\pi}_\ell(a)$ for 27 fuzzy states.

1) Fused Q-Value: The effective Q-value at state s is computed as a fuzzy-weighted combination:

$$\mathrm{FQ}(s,a) = \sum_{\ell=1}^{27} \psi_\ell(s) \cdot Q_\ell(s,a). \qquad (17)$$

2) Q-Update: For each active fuzzy state ℓ (where $\psi_\ell > 0$), the corresponding entry is updated toward the fuzzy-weighted Bellman target:

$$Q_\ell(s,a) \leftarrow (1-\alpha)\,Q_\ell(s,a) + \alpha\left[r + \gamma \max_{a'} \mathrm{FQ}(s',a')\right] \qquad (18)$$

where $\gamma = 0.9$ is the discount factor and α is a visit-count-adaptive learning rate,

$$\alpha = \min\left(1,\ \alpha_0\left(1 + \frac{3}{1 + 0.1\,n(s,a)}\right)\right), \qquad (19)$$

with base rate $\alpha_0 = 0.01$ and $n(s,a)$ the visit count of action a in discrete state s; this boosts learning for rarely-visited state–action pairs and anneals toward α_0 as experience accumulates.

3) WoLF-PHC Policy Update: The win-or-learn-fast mechanism adapts the learning speed based on performance:

$$\xi = \begin{cases} \xi_{\mathrm{win}} = 0.02 & \text{if } \mathbb{E}_\pi[Q] > \mathbb{E}_{\bar{\pi}}[Q] \\ \xi_{\mathrm{loss}} = 0.08 & \text{otherwise} \end{cases} \qquad (20)$$

The policy is updated with step size $\delta = \xi \cdot \psi_\ell$:

$$\pi_\ell(a^*) \leftarrow \pi_\ell(a^*) + \delta \qquad (21)$$

$$\pi_\ell(a) \leftarrow \pi_\ell(a) - \frac{\delta}{|\mathcal{A}| - 1}, \quad \forall a \neq a^* \qquad (22)$$

where $a^* = \arg\max_a \mathrm{FQ}(s,a)$.

The average policy $\bar{\pi}_\ell$ uses exponential smoothing. During evaluation, actions are sampled from an adaptive Boltzmann softmax over $\mathrm{FQ}(s,\cdot)$, with temperature scaled to the Q-value spread so that probability concentrates on the best action when its margin is clear and spreads out when values are close; a small uniform-mixing floor (3%) is additionally applied to every action so that the jammer can never fully predict the agent's behavior.

E. Reward Function

The reward balances throughput, power efficiency, and QoS:

$$r = \sum_{k=1}^{K} \log_2(1+\mathrm{SINR}_k) - \lambda_1 \frac{\sum_k P_k}{P_{\max}} - \lambda_2 \cdot \frac{1}{K} \sum_{k=1}^{K} \frac{[\gamma_{\min} - \bar{\gamma}_k]^+}{|\gamma_{\min}|} \qquad (23)$$

where $[\cdot]^+ \triangleq \max(0,\cdot)$ (clipped to a maximum deficit of 30 dB), $\bar{\gamma}_k \triangleq \frac{1}{M_{\mathrm{sc}}} \sum_{m=1}^{M_{\mathrm{sc}}} 10\log_{10} \mathrm{SINR}_k[m]$ is the subcarrier-averaged SINR of user k *in dB* (distinct from the linear-scale

SINR_k used in the rate term above), and $\lambda_1 = 0.5$, $\lambda_2 = 1.5$. This continuous SINR-deficit penalty replaces a binary outage indicator: since it scales smoothly with how far a user is below $\gamma_{\min}$, it avoids the flat reward landscape that a binary penalty produces when all users are simultaneously in outage.

F. Smart Jammer Model

The adaptive jammer monitors SINR feedback and policy predictability. Prior to temporal smoothing, its target power toward user k is

$$\tilde{P}_{J,k}\,[\mathrm{dBm}] = P_{J,\min} + 0.25\,\mathrm{clip}(\mathrm{SINR}_k^{\mathrm{prev}} - 5, 0, 20) + 18\eta + n_J \tag{24}$$

where $P_{J,\min} = 15\,\mathrm{dBm}$, $\eta \in [0, 1]$ is a predictability score from the last 20 actions, and $n_J \sim \mathcal{N}(0, 2.25)$ dB. To avoid abrupt power swings that destabilize training, the applied power is exponentially smoothed across steps,

$$P_{J,k}[t] = 0.7\,P_{J,k}[t-1] + 0.3\,\tilde{P}_{J,k}[t]. \tag{25}$$

The precoder interpolates between isotropic and user-directed jamming:

$$\mathbf{z}_k = (1 - w_{\mathrm{align}})\mathbf{z}_{\mathrm{random}} + w_{\mathrm{align}} \frac{\mathbf{h}_{J,k}}{\|\mathbf{h}_{J,k}\|} \tag{26}$$

with $w_{\mathrm{align}} = \mathrm{clip}(2(\eta - 0.5), 0, 1)$.

G. Baseline Methods

Baselines:

- **Classical Q-learning [12]** with ε-greedy updates.
- **Fast Q-learning [9]** with visit-count-boosted learning rate.
- **DQN [13]** with replay and target network.
- **AO baseline [6]** with fixed channel-proportional power.
- **No-IRS** lower bound.

These baselines were chosen to span foundational tabular RL, a representative IRS-anti-jamming fast-RL method, deep RL, and deterministic AO optimization, providing methodologically diverse and widely-used reference points in the IRS anti-jamming literature. ~~Comparison against more recent (2023–2025) model-based or transformer-based anti-jamming schemes is left for future evaluation.~~

IV. Simulation Results

A. Simulation Setup

Simulations are implemented in Python (NumPy/PyTorch). The THz system uses $f_c = 100\,\mathrm{GHz}$, $B = 10\,\mathrm{GHz}$, $N = 64$, $N_{\mathrm{RF}} = 8$, $M = 256$, $K = 4$, and $M_{\mathrm{sc}} = 64$. Users are randomly placed in $[30, 120] \times [0, 80]$ m, while the jammer follows a bounded random walk in $[40, 100] \times [0, 80]$ m. Each method is trained for 600 episodes (20 steps per episode) over 5 independent random seeds; all parameter sweep figures are produced from full real-data runs (340 training/evaluation tasks total) without any curve fitting or interpolation. Key parameters are listed in Table II.

TABLE II
Simulation Parameters

Parameter	Symbol	Value
Center frequency	f_c	100 GHz
Bandwidth	B	10 GHz
BS antennas	N	64
RF chains	N_{RF}	8
IRS elements	M	256 (16×16)
IRS sub-arrays	Q	64 (8×8)
OFDM subcarriers	M_{sc}	64
Users	K	4
Jammer antennas	N_J	2
Max TX power	$P_{\max}$	40 dBm
SINR threshold	$\gamma_{\min}$	5 dB
Noise figure	F_{NF}	10 dB
RL actions	$\lvert\mathcal{A}\rvert$	42
Training episodes	E	600
Seeds	—	5

TABLE III
Evaluation Performance (Mean $\pm$ Std over 5 Seeds)

Method	Rate (bits/s/Hz)	Protection (%)
No IRS	5.48	18.8
AO Baseline [6]	7.89 ± 0.78	43.2 ± 7.7
Classical Q-learning	8.25 ± 1.28	49.0 ± 14.4
Fast Q-learning [9]	8.05 ± 1.68	45.5 ± 18.7
DQN [13]	8.23 ± 2.40	49.1 ± 25.7
Fuzzy WoLF-PHC (proposed)	$\mathbf{11.87 \pm 0.55}$	$\mathbf{81.6 \pm 3.9}$

B. Overall Performance Comparison

Table III and Fig. 1 summarize results over five seeds. Fuzzy WoLF-PHC achieves 11.87 bits/s/Hz and 81.6% protection, improving over the next-best baseline by 44% (rate) and 66% (protection), while also yielding the lowest variance (± 0.55, $\pm 3.9\%$). This indicates that mixed-strategy adaptation is both stronger and more stable under adversarial dynamics. The marginal standard deviations of the tabular and deep-RL baselines are large (e.g., DQN's protection varies by $\pm 25.7\%$), largely reflecting across-seed differences in topology and jammer difficulty rather than algorithmic instability: the four RL-trained methods share identical per-seed channel/jammer realizations at evaluation, making a paired comparison possible, and Fuzzy WoLF-PHC outperforms Classical Q-learning, Fast Q-learning, and DQN in every one of the 5 seeds for both rate and protection (30 of 30 paired comparisons). It additionally exceeds the independently-seeded AO baseline in all 5 seed replicates for both metrics.

C. Effect of Maximum Transmit Power

Fig. 2 shows rate/protection versus $P_{\max} \in \{25, 32, 40\}$ dBm. WoLF-PHC rises from 1.72 to 11.87 bits/s/Hz and from 0% to 81.6%, showing stronger scaling with available power than deterministic AO.

D. Effect of RIS Array Size

Fig. 3 studies $N_{\mathrm{RIS}} \in \{16, 64, 144, 256\}$. WoLF-PHC improves from 11.47 to 11.87 bits/s/Hz and from 79.2% to 81.6%, confirming consistent gains with larger IRS size, with expected saturation at high element counts.

Fig. 1. Evaluation comparison: (a) system rate and (b) SINR protection (mean $\pm$ std, 5 seeds). Fuzzy WoLF-PHC (blue) achieves the highest performance with the smallest variance across both metrics.

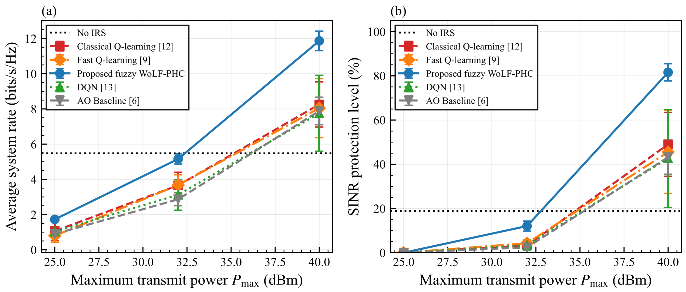

Fig. 2. (a) System rate and (b) SINR protection vs. maximum transmit power $P_{\max}$. WoLF-PHC scales most effectively with power, opening a gap over deterministic and tabular RL baselines at 40 dBm.

E. Effect of SINR Threshold

Fig. 4 evaluates $\gamma_{\min} \in \{3, 5, 10, 15, 20\}$ dB. WoLF-PHC maintains the best trade-off across thresholds: 91.8% protection at 3 dB, 81.6% at 5 dB, and 31.4% at 10 dB. All methods collapse at 20 dB, indicating a hard operating limit under the considered jammer.

F. Robustness to Jammer Power

Fig. 5 shows a regime transition versus $P_J \in \{5, 10, 15, 20\}$ dBm. AO is competitive at low jammer power, but WoLF-PHC dominates for $P_J \geq 15$ dBm and reaches 70.2% protection at 20 dBm, highlighting the value of mixed-strategy unpredictability.

G. Discussion

Results confirm three points: WoLF-PHC gives the best and most stable rate/protection trade-off; robustness gains are largest in medium/high-jammer regimes; and larger IRS sizes provide consistent, though saturating, improvements.

V. Conclusion

This paper proposed a hybrid RL–AO framework for IRS-assisted anti-jamming in wideband THz systems, jointly optimizing transmit power allocation, SPDP-RIS phase control, and hybrid beamforming under adaptive jamming. A fuzzy WoLF-PHC-based controller was developed to enable robust mixed-strategy power allocation in dynamic adversarial environments.

Simulation results demonstrate that the proposed method achieves a system rate of 11.87 bits/s/Hz and an SINR protection level of 81.6%, outperforming the best baseline by 44% in throughput and 66% in protection, while maintaining the lowest variance across independent runs. Extensive parameter studies further confirm the robustness of the proposed framework under varying transmit power, IRS size, SINR thresholds, and jammer strength, revealing a clear operating regime in which learning-based strategies significantly outperform deterministic optimization methods.

Future work will extend this framework to imperfect CSI scenarios, multi-cell networks, and near-field THz propagation with practical IRS hardware impairments.

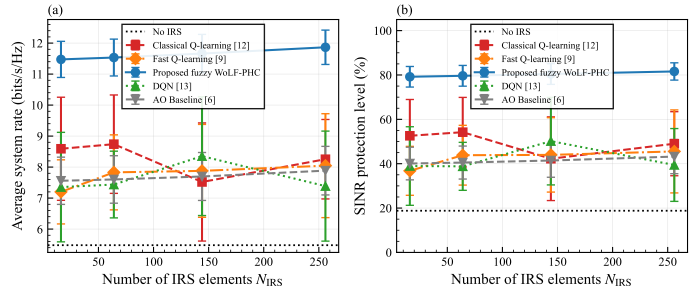

Fig. 3. (a) System rate and (b) SINR protection vs. IRS array size N_{RIS}. Performance grows with array size but saturates near $N_{\mathrm{RIS}} = 256$; WoLF-PHC retains a consistent advantage at all scales.

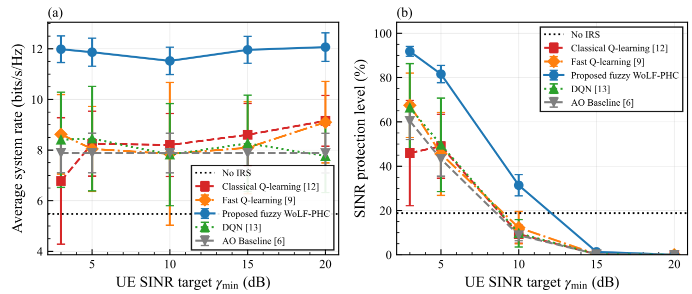

Fig. 4. (a) System rate and (b) SINR protection vs. UE SINR target $\gamma_{\min}$. WoLF-PHC shows the most graceful degradation with increasing SINR threshold, retaining 31.4% protection at 10 dB.

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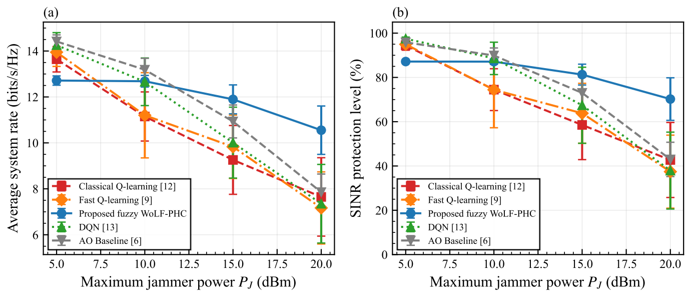

Fig. 5. (a) System rate and (b) SINR protection vs. maximum jammer power P_J. A clear regime transition occurs near 10–15 dBm: WoLF-PHC's advantage grows with jammer power, while deterministic AO degrades sharply.

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